HOME WORK ASSIGNMENT: 1

Ans: to the question: 1

(a) There are several ways to represent or store graphs, each with its advantages and disadvantages. Let us look at the ways to represent or store a graph in memory. Here are some ways:

→ **Adjacency matrices:** For a graph with $|V|$ vertices an adjacency matrix is a $|V| \times |V|$, where the entry in row i and column j is 1 if only if the edge (i, j) is in the graph.

→ **Adjacency list:** Representing a graph with adjacency list combines adjacency matrices with list. For each vertex i, store an array of the vertices adjacent to it. For an undirected graph adj. list contain $2|E|$ elements and for a directed graph it contains a total of $|E|$.

→ **Edge list:** Edge list is a simple way to represent a graph. We just need an array of two vertex or an array of objects containing the vertex number of vertices.

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Let us use Depth first Search for the following graph, it will be E, C, A, G, B, D, F

Representation of the graph by adjacency matrix:

	A	B	C	D	E	F	G
A	0	1	0	0	0	0	0
B	0	0	1	0	0	0	1
C	0	0	0	0	1	0	0
D	0	0	1	0	1	0	0
E	0	0	0	0	0	0	0
F	0	0	1	0	1	0	0
G	1	0	1	0	0	0	0

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→ Tree Edges:

(b) Starting from node 'A', there is edge 'B' because 'B' is athe child of 'A' and 'C' is the child of 'B'. So there is a relation among these nodes like a tree has with its braimches.

→ Forward Edges: There are no forward edges of node 'A' because 'A' is not connected with its tree edges edge.

→ Backward Edges: 'G' is the backward edge for node 'A' because it is a child edge of 'B' node and there is a opposite relation between 'A' and 'G'.

→ Cross Edge: There are no cross edges for node 'A' but we see there is a cross connection with 'F', 'D', 'G' beacuse the are im different trees.

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After applying BFS on that graph

A → B → G → C → E → D → F

Ans: to the question: 2

(a) Few Real life applications using some Graph Algorithms.

→ Social graph draw edges among people, places and things you interact with online. While connecting friends on social site, each user is a vertex and their connection make edges. [MST]

→ We use Maps to find location from map which refers to vertices of graph. Distances between location refers to edges. It is also used in telephone network Also it is used in GPS. [Dijkstra]

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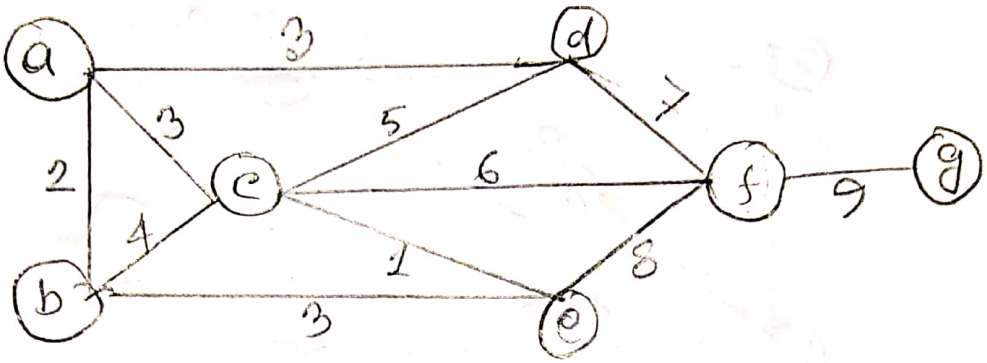
→ There are algorithm which are used for making cartography, calculating mathematical terms, for creating Mechanical and electrical equipment and study of Molecular Chemistry. For these programmer use [Kruskal & prims]

→ Searching on Google for webpages where pages are linked to each other. So every page is a vertex and link between them is an edge. [DFS]

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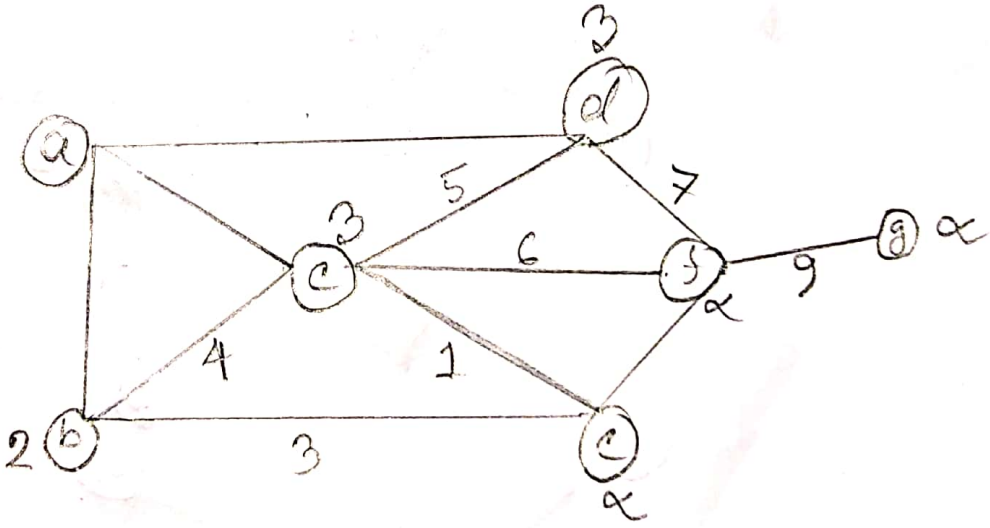
(b)

⇒



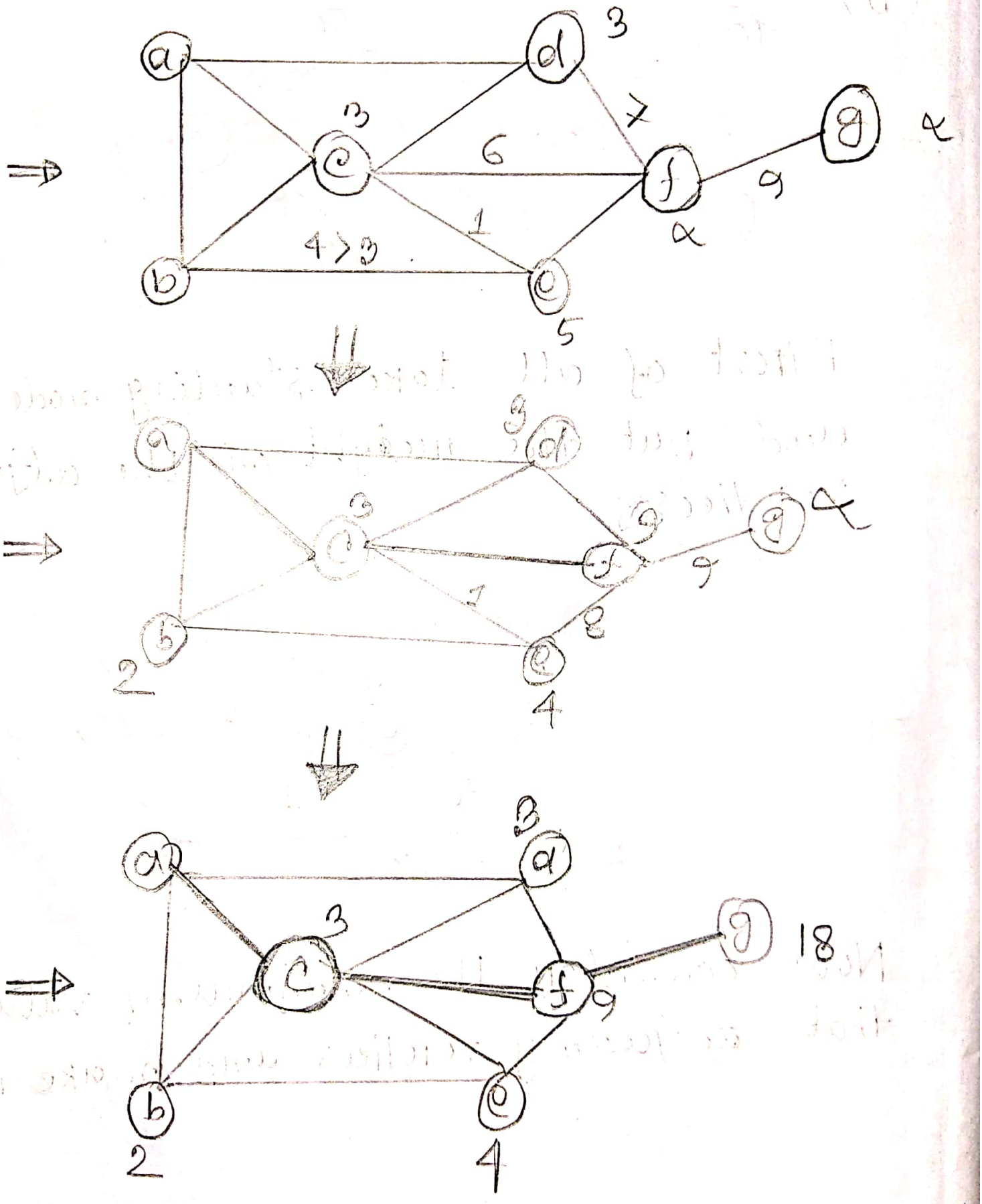
First of all take starting node 'a' and put the weight of the adjacency vertices.

⇒



Now consider the minimum value of that adjacency vertices and make relation

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Here we can see the shortest path from 'A' to 'G' is 18 //

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